

CHECK_DISTANCE

Overview

check_distance verifies that a **start_event** is followed by a **test_event**, and the clock distance between the **start_event** and the **test_event** falls within a predefined distance range. Not before a minimum of **min_clks** and not later a maximum of **max_clks** clock cycles from the start event, it's test_event must be true.

Check_distance supports reentrancy or overlapping sequences. It means that any number of **start_events** may precede the number of **test_events**, and the distance is checked for each pair of corresponding start and test events.

Check_distance allows users to associate the start and test events with a signal level, rising edge, falling edge or signal change.

Check_distance performs an end-of-simulation check to make sure there are no "unacknowledged" start events.

Syntax

```
check_distance #(min_clks,max_clks,start_type,test_type) InstName (clk,  
reset_n, start_event, test_event);
```

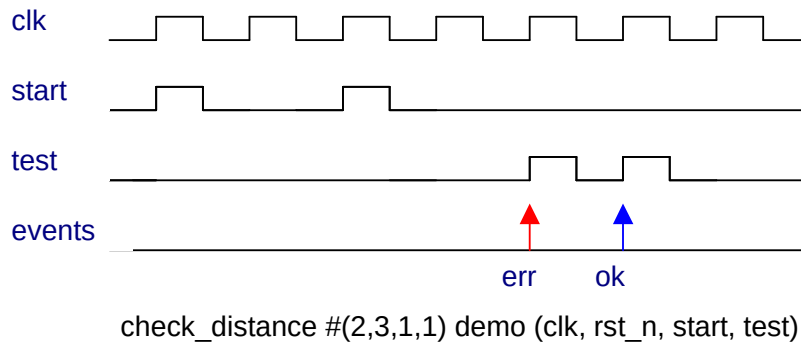
Parameters	
min_clks	Minimum distance in clock cycles between the start and test events
max_clks	Maximum distance in clock cycles between the start and test events
start_type, test_type	Enumerated types select transition types for the start and test events: 0 – signal level 1 – rising edge 2 – falling edge 3 – any change
Interface	
clk	Clock input
reset_n	Active-low reset
start_event	1-bit signal defining a start event
test_event	1-bit signal defining a test event
Access Registers	
distance	Monitors the actual distance in clock cycles between the start and test events. Can be used for statistics monitoring with a vc_stat component.
Access Events	
err, ok	Signals about failure or success. Can be monitored externally for the test control purposes.
Defines	
TOP_FINISH	Monitors the "top.finish" event, which signals to all verification components about the end of simulation. In case of an "unacknowledged" start event(s), it issues an error to using "err" access event as well as an error display message.
ASSERT_FATAL	In case of an assertion failure, it issues a call to the top-level "top.error" task, which performs a test case exit under error conditions. Note: In order to use defines, the following rules apply: 1. Name of the top-level TestBench module is "top". 2. Top level contains an event called "finish". This event has to fire every time before a simulation exit. 3. Top level contains a task called "error". This task performs the exit-under-error procedure for the whole TestBench.

Usage

Check_distance is a powerful verification component, replacing and enhancing the following OVL assertion monitors:

- Assert_change
- Assert_frame
- Assert_next

As it has been pointed out earlier, **check_distance** supports any number of overlapped sequences. The following figure demonstrates the way this process works:



Verilog Example

```
`define TOP_FINISH 1
module top;

event finish;
reg clk, rst_n;
reg req, ack;
initial clk = 1;
always #5 clk <= ~clk;

task finish_err;
begin
// Implements error exit procedure
end
endtask

check_distance #(3,6,0,0) d1 (clk, rst_n, req, ack);

initial begin
    req = 0; ack = 0; rst_n = 0;
    #20 rst_n = 0;
    #10 req = 1;    #10 req = 0;
    #10 req = 1;    #10 req = 0;
    #10 ack = 1; #10 ack = 0;
    #10 ack = 1; #10 ack = 0;
    #100 -> finish;
    #1 $finish;
end

endmodule
```